

Package ‘GR2MSemiDistr’

August 12, 2025

Type Package

Title Semi-Distributed GR2M Model for Large-Sample Hydrological Applications

Version 5.0

Author Harold Llauca <hllauca@senamhi.gob.pe>

Maintainer Harold Llauca <hllauca@senamhi.gob.pe>

Description Implements a semi-distributed adaptation of the GR2M monthly hydrological model for large-sample and operational hydrological modeling. The model uses regional parameters and subbasin-level forcing data to simulate water balance components such as precipitation, evapotranspiration, runoff, and discharge. It supports warm-up periods, initial states, output saving, regional parameter correction, and performance evaluation through various objective functions. The routing scheme is based on subbasin structure and spatial relationships derived from a DEM using WhiteboxTools. Designed for flexible integration in research, water resource assessments, and operational hydrological systems in complex basins.

License GPL (>= 2)

Encoding UTF-8

Depends R (>= 4.0)

Imports airGR, hydroGOF, Matrix, terra, lubridate, tictoc, sf, rtop, whitebox, exactextractr

RoxygenNote 7.3.2

URL <https://github.com/hllauca/GR2MSemiDistr>

BugReports <https://github.com/hllauca/GR2MSemiDistr/issues>

RemoteType github

RemoteHost api.github.com

RemoteRepo GR2MSemiDistr

RemoteUsername hllauca

RemoteRef HEAD

RemoteSha 7ae6bec7b7c03950331afb78b238d79f511e674d

GithubRepo GR2MSemiDistr

GithubUsername hllauca

GithubRef HEAD

GithubSHA1 7ae6bec7b7c03950331afb78b238d79f511e674d
NeedsCompilation no

Contents

Create_Forcing_Inputs	2
Load_example_data	4
Optim_GR2MSemiDistr	4
Routing_GR2MSemiDistr	7
Run_GR2MSemiDistr	9

Index	12
--------------	-----------

Create_Forcing_Inputs	<i>Extract and prepare model input data in the airGR format (DatesR, P, E, and optionally Q) using gridded monthly precipitation and evapotranspiration.</i>
-----------------------	--

Description

This function extracts spatially averaged monthly precipitation and potential evapotranspiration values for each subbasin, and optionally includes observed streamflow data, to produce a formatted data frame compatible with the airGR hydrological modeling framework.

Usage

```
Create_Forcing_Inputs(  
  Subbasins,  
  Precip,  
  PotEvap,  
  Qobs = NULL,  
  DateIni,  
  DateEnd,  
  IniGrids = "01/1981",  
  Save = FALSE,  
  Update = FALSE,  
  Members = NULL  
)
```

Arguments

Subbasins	A SpatVector object containing the subbasins' geometries. Must include the following attributes: 'COMID' (unique identifier), and 'Region' (string code for the hydro-climatic region).
Precip	A SpatRaster object with monthly precipitation data in [mm/month].
PotEvap	A SpatRaster object with monthly potential evapotranspiration data in [mm/month].
Qobs	Optional. A vector with observed streamflow in [m³/s] at the basin outlet.
DateIni	Start date of the output time series in 'mm/yyyy' format.
DateEnd	End date of the output time series in 'mm/yyyy' format.

IniGrids	Initial date of the gridded precipitation and evapotranspiration datasets in mm/yyyy format.
Save	Logical. If TRUE, saves the resulting data frame to a .txt file in the './Inputs' directory. Default is FALSE.
Update	Logical. If TRUE, returns only the last month of data (useful for operational updates). Default is FALSE.
Members	Optional. Integer indicating the number of ensemble members for streamflow forecasting. If provided, the date vector will be repeated accordingly. Default is NULL.

Value

A data frame containing the model inputs in the airGR format, with columns:

DatesR Monthly dates

P_x Mean precipitation for subbasin x, where x is the COMID

E_x Mean potential evapotranspiration for subbasin x, where x is the COMID

Q Observed streamflow [m³/s], if provided

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

Examples

```
library(GR2MSemiDistr)

# Define simulation period and grid start date
start_date <- "01/1981"
end_date   <- "12/2016"
ini_grids  <- "01/1981"

# Create input data for the model
model_inputs <- Create_Forcing_Inputs(
  Subbasins = cat,
  Precip = grid_pr,
  PotEvap = grid_pe,
  Qobs = qobs,
  DateIni = start_date,
  DateEnd = end_date,
  IniGrids = ini_grids,
)

# Create figures to visualize the inputs
dates <- as.Date(model_inputs$DatesR)

# Plot average precipitation across subbasins
P_vars <- grep("^P_", names(model_inputs), value = TRUE)
avg_P <- rowMeans(model_inputs[, P_vars], na.rm = TRUE)
plot(dates, avg_P, type = "l", col = "blue", lwd = 1.5,
     ylab = "Precipitation [mm/month]", xlab = "Date",
     main = "Mean Precipitation across Subbasins")
```

```
# Plot average potential evapotranspiration across subbasins
E_vars <- grep("^E_", names(model_inputs), value = TRUE)
avg_E <- rowMeans(model_inputs[, E_vars], na.rm = TRUE)
plot(dates, avg_E, type = "l", col = "orange", lwd = 1.5,
     ylab = "PET [mm/month]", xlab = "Date",
     main = "Mean Potential Evapotranspiration across Subbasins")

# Plot observed streamflow at outlet (if available)
if ("Q" %in% names(model_inputs)) {
  plot(dates, model_inputs$Q, type = "l", col = "darkgreen", lwd = 1.5,
       ylab = "Observed Streamflow [m³/s]", xlab = "Date",
       main = "Observed Streamflow at Outlet")
}
```

Load_example_data	<i>Load example hydrological data included in the package</i>
-------------------	---

Description

Loads spatial and tabular example data (raster and vector) from 'inst/extdata/'.

Usage

```
Load_example_data()
```

Value

A list with objects: 'cat', 'dem', 'grid_pr', 'grid_pe', 'qobs'

Examples

```
# Load example data

data <- GR2MSemiDistr::Load_example_data()
names(data)
cat <- data$cat
dem <- data$dem
qobs <- data$qobs
grid_pr <- data$grid_pr
grid_pe <- data$grid_pe
```

Optim_GR2MSemiDistr	<i>Parameter optimization for a semi-distributed GR2M model using the SCE-UA algorithm.</i>
---------------------	---

Description

Optimizes the GR2M monthly water balance model parameters (X1, X2, fp, fe) for multiple calibration regions using the SCE-UA algorithm. The model is run in a semi-distributed setup over multiple subbasins grouped by region. Users can select from various objective functions (e.g., NSE, KGE, RMSE, and linear combinations), including composite metrics. Parameter bounds must be provided, and regions can be excluded from calibration.

Usage

```

Optim_GR2MSemiDistr(
  Data,
  Subbasins,
  RunIni,
  RunEnd,
  Parameters,
  Parameters.Min = c(100, 0.1, 0.8, 0.8),
  Parameters.Max = c(2000, 10, 1.2, 1.2),
  Optimization = "OF1",
  Max.Functions = 1000,
  WarmUp = 36,
  No.Optim = NULL,
  w1 = 0.6,
  w2 = 0.3,
  w3 = 0.2
)

```

Arguments

Data	A data frame of model input data in the airGR format, as produced by <code>Create_Forcing_Inputs</code> . It must include columns: <code>DatesR</code> , <code>P_1</code> to <code>P_n</code> , <code>E_1</code> to <code>E_n</code> , and <code>Q</code> (used for calibration).
Subbasins	A <code>SpatVector</code> object containing the geometries of subbasins. It must include attributes "COMID" (unique subbasin ID) and "Region" (region name/code).
RunIni	Simulation start date in the format "mm/yyyy".
RunEnd	Simulation end date in the format "mm/yyyy".
Parameters	A data frame containing GR2M model parameters and correction factors per region. Must have columns: <code>Region</code> , <code>X1</code> , <code>X2</code> , <code>fp</code> , <code>fe</code> .
Parameters.Min	Numeric vector of length 4, specifying lower bounds for optimization in the following order: <code>c(X1, X2, fp, fe)</code> .
Parameters.Max	Numeric vector of length 4, specifying upper bounds for optimization in the same order as 'Parameters.Min'.
Optimization	Character. Objective function to optimize. Available options include: "OF1" Kling-Gupta Efficiency (KGE). "OF2" Nash-Sutcliffe Efficiency (NSE). "OF3" NSE applied to log-transformed flows (log-NSE, Pushpalatha 2012). "OF4" Root Mean Square Error (RMSE). "OF5" Percent Bias (PBIAS). "OF6" FDC Bias (FDC Bias), based on flow duration curves. "OF7" Pearson correlation coefficient (r). "OF8" Composite objective: $w1 \times KGE + w2 \times NSE - w3 \times RMSE$. "OF9" Composite objective: $w1 \times KGE + w2 \times \text{log-NSE} - w3 \times RMSE$. "OF10" Composite objective: $w1 \times KGE.km + w2 \times KGE.If - w3 \times RMSE$.
Max.Functions	Integer. Maximum number of function evaluations in the optimization. Default is 1500.
No.Optim	Character vector (optional). Region names to exclude from the optimization (parameters will be kept fixed).

w1	Numeric. Weight for the first component in composite objective functions (OF8, OF9, OF10). Default is 0.6.
w2	Numeric. Weight for the second component in composite objective functions. Default is 0.3.
w3	Numeric. Weight for the third component in composite objective functions. Default is 0.2.

Value

A list with the following elements:

Parameters Data frame of optimized parameters per region (columns: Region, X1, X2, fp, fe).

OF Final value of the selected objective function.

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

Examples

```
library(GR2MSemiDistr)

# Define initial parameters for each region
param_init <- data.frame(
  Region = unique(cat$Region),
  X1 = rep(500, length(unique(cat$Region))),
  X2 = rep(1.5, length(unique(cat$Region))),
  fp = rep(1.0, length(unique(cat$Region))),
  fe = rep(1.0, length(unique(cat$Region)))
)

# Optimize parameters using the KGE criterion (OF1)
result <- Optim_GR2MSemiDistr(
  Data = model_inputs,
  Subbasins = cat,
  RunIni = "01/1981",
  RunEnd = "12/2005",
  Parameters = param_init,
  Optimization = "OF1" # KGE
)

# Extract results
best_params <- result$Parameters
final_score <- result$OF

# Plot observed vs simulated discharge at the outlet
model <- Run_GR2MSemiDistr(
  Data = model_inputs,
  Subbasins = cat,
  RunIni = "01/1981",
  RunEnd = "12/2005",
  Parameters = best_params,
)
```

```

if (!is.null(model$SINK)) {
  par(mfrow = c(1, 1))
  plot(dates, model$SINK$sim, type = "l", col = "blue",
        xlab = "Date", ylab = "Discharge [m³/s]",
        lines(dates, model$SINK$obs, col = "red", type='o'))
  legend("topright", legend = c("Simulated", "Observed"),
        col = c("blue", "red"), lty = c(1, 2), lwd = 2, bty = "n")
}

```

Routing_GR2MSemiDistr *Routing of simulated discharges for each subbasin using a weighted flow-accumulation network*

Description

Routes monthly simulated streamflows from a semi-distributed GR2M model through a single-downstream (D8) river network. The routing network is derived from a Digital Elevation Model (DEM) with WhiteboxTools, or provided directly via a precomputed transfer matrix. Each sub-basin's outflow is accumulated downstream using the reachability (accumulation) matrix built from the transfer matrix.

Usage

```

Routing_GR2MSemiDistr(
  Model,
  Subbasins,
  Dem,
  RouIni = NULL,
  RouEnd = NULL,
  TransferMatrix = NULL,
  res = 500,
  Save = FALSE,
  Update = FALSE
)

```

Arguments

Model	List with GR2M results, typically from Run_GR2MSemiDistr. Must include: QS Numeric matrix of simulated streamflows [m³/s]; columns are subbasins (COMID). Dates Vector of dates corresponding to the rows of QS.
Subbasins	A SpatVector of subbasin boundaries. Must contain attributes: COMID Integer ID of each subbasin (unique). Region Region code or name.
Dem	A SpatRaster DEM used to derive the routing network (required unless TransferMatrix is supplied). If the DEM is in lon/lat, res is converted to degrees.
RouIni	Character. Start date for routing in "mm/YYYY" format. If NULL, uses the beginning of Model\$Dates.

RouEnd	Character. End date for routing in "mm/YYYY" format. If NULL, uses the end of Model\$Dates.
TransferMatrix	Optional numeric square matrix (n x n) encoding direct downstream connectivity (rows = destinations, columns = sources). If provided, Dem and WhiteboxTools are not used. Row/column names (COMID) must match Subbasins\$COMID order; if missing, they are set from COMID.
res	Numeric. Target DEM resolution in meters for resampling (default 500). Converted to degrees when Dem is in lon/lat.
Save	Logical. If TRUE, writes routed discharges (QR) as a tabular .txt file into "./Outputs". Default FALSE.
Update	Logical. If TRUE, removes the previous month's output before saving the new one (useful for incremental/operational runs). Default FALSE.

Details

The transfer matrix T encodes single-downstream connectivity between subbasins: $T[i, j] = 1$ if subbasin j drains directly to i , and 0 otherwise. The accumulation matrix is $A = I + T + T^2 + \dots$ (truncated within $N-1$ steps for a DAG), so that routed flows are computed as $QR = QS$. QR are simulated outflows per subbasin (columns) and time steps (rows).

When deriving T from a DEM, D8 flow directions and flow accumulation (FAC) are computed with WhiteboxTools. For each subbasin, the unique downstream neighbor is selected from boundary cells by the candidate with the highest FAC. Subbasin rasterization uses `touches = FALSE` to avoid spurious edge assignments.

Value

A list with:

QR Base matrix of routed streamflows [m^3/s]; rows = time, columns = COMID.

Dates Vector of routing dates used for QR.

COMID Character vector of subbasin IDs (column order of QR).

TransferMatrix Base matrix used for routing; $T[i, j] = 1$ if $j \rightarrow i$.

AccumMatrix Base matrix A used to accumulate upstream contributions.

References

Llaucha H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

See Also

Run_GR2MSemiDistr

Examples

```
library(GR2MSemiDistr)

# Run routing using DEM and WhiteboxTools
routed <- Routing_GR2MSemiDistr(
  Model = model,
  Subbasins = cat,
```

```

    Dem = dem,
    RouIni = "01/1981",
    RouEnd = "12/2016",
  )

  # Select COMID of the subbasin to plot
  idx <- which(model$COMID == model$COMID[1]) # Change index as needed
  dates <- as.Date(model$Dates)

  plot(dates, routed$QR[, idx], type = "l", col = "darkblue", lwd = 2,
       xlab = "Date", ylab = "Discharge [m³/s]",
       main = paste("Routed Discharge - Subbasin", routed$COMID[idx]))

```

Run_GR2MSemiDistr	<i>Run the GR2M model for multiple subbasins</i>
-------------------	--

Description

This function executes the GR2M monthly water balance model across multiple subbasins using a semi-distributed approach. It applies region-specific parameters and correction factors for precipitation and evapotranspiration, and optionally saves the results as text files.

Usage

```

Run_GR2MSemiDistr(
  Data,
  Subbasins,
  RunIni,
  RunEnd,
  Parameters,
  WarmUp = NULL,
  IniState = NULL,
  Save = FALSE,
  Update = FALSE
)

```

Arguments

Data	A data frame of model input data in the airGR format, as produced by Create_Forcing_Inputs. It must include columns: DatesR, P_1 to P_n, E_1 to E_n, and optionally Q.
Subbasins	A SpatVector object containing the geometries of subbasins. It must include attributes 'COMID' (unique subbasin ID) and 'Region' (region name/code).
RunIni	Simulation start date in the format 'mm/yyyy'.
RunEnd	Simulation end date in the format 'mm/yyyy'.
Parameters	A data frame containing GR2M model parameters and correction factors per region. Must have columns: Region, X1, X2, fp, fe.
WarmUp	Optional number of months to discard from the beginning of the simulation as warm-up. Default is NULL.
IniState	Optional list of initial states for each subbasin. Default is NULL.

Save	Logical. If TRUE, output time series will be saved as text files in the "Outputs/" folder.
Update	Logical. If TRUE and Save = TRUE, it will remove previous month's output files before saving the current ones.

Value

A list with the following elements:

PR Matrix of total precipitation [mm/month] for each subbasin

AE Matrix of actual evapotranspiration [mm/month] for each subbasin

SM Matrix of production store level [mm/month] for each subbasin

SM Matrix of percolation [mm/month] for each subbasin

RU Matrix of runoff [mm/month] for each subbasin

QS Matrix of flow [m3/s] (not routed) for each subbasin

Dates Date vector for the simulation period

COMID Vector of subbasin COMIDs

EndState List of final state variables for each subbasin

SINK Optional. Data frame of simulated and observed outlet discharge, if observed Q is provided

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

Examples

```
library(GR2MSemiDistr)

# Define region parameters
param_df <- data.frame(
  Region = unique(cat$Region),
  X1 = 2000,
  X2 = 1.8,
  fp = 0.8,
  fe = 0.8
)

# Run the semi-distributed GR2M model
model <- Run_GR2MSemiDistr(
  Data = model_inputs,
  Subbasins = cat,
  RunIni = "01/1981",
  RunEnd = "12/2016",
  Parameters = param_df,
)

# Select COMID of the subbasin to plot
target_comid <- model$COMID[1] # Change index as needed
idx <- which(model$COMID == target_comid)
dates <- as.Date(model$Dates)
```

```

# Plot model outputs for selected subbasin
par(mfrow = c(3, 2), mar = c(4, 4, 3, 1))

plot(dates, model$PR[, idx], type = "l", col = "blue",
     main = paste("Precipitation (PR) - COMID", target_comid),
     xlab = "Date", ylab = "mm/month")

plot(dates, model$AE[, idx], type = "l", col = "orange",
     main = paste("Actual Evapotranspiration (AE) - COMID", target_comid),
     xlab = "Date", ylab = "mm/month")

plot(dates, model$SM[, idx], type = "l", col = "green4",
     main = paste("Production Store (SM) - COMID", target_comid),
     xlab = "Date", ylab = "mm/month")

plot(dates, model$PC[, idx], type = "l", col = "purple",
     main = paste("Percolation (PC) - COMID", target_comid),
     xlab = "Date", ylab = "mm/month")

plot(dates, model$RU[, idx], type = "l", col = "darkred",
     main = paste("Runoff (RU) - COMID", target_comid),
     xlab = "Date", ylab = "mm/month")

plot(dates, model$QS[, idx], type = "l", col = "blue",
     main = paste("Simulated Discharge (QS) - COMID", target_comid),
     xlab = "Date", ylab = "m³/s")

# Optional: compare simulated vs observed outlet discharge
if (!is.null(model$SINK)) {
  par(mfrow = c(1, 1))
  plot(dates, model$SINK$sim, type = "l", col = "blue",
       xlab = "Date", ylab = "Discharge [m³/s]",
       lines(dates, model$SINK$obs, col = "red", type='o'))
  legend("topright", legend = c("Simulated", "Observed"),
        col = c("blue", "red"), lty = c(1, 2), lwd = 2, bty = "n")
}

```

Index

Create_Forcing_Inputs, [2](#)

Load_example_data, [4](#)

Optim_GR2MSemiDistr, [4](#)

Routing_GR2MSemiDistr, [7](#)

Run_GR2MSemiDistr, [9](#)